Assignment 0522

REQUIREMENT: Please solve the following problems exclusively using pointer operations. If you use standard array subscript syntax (e.g., `a[0]` , `a[i][j]`) anywhere in your solution, you will receive zero points for that problem.

1

In learned video compression, 3×3 matrices are frequently used as convolutional filters to process spatial features. Write a C program that calculates the determinant of a given 3×3 filter matrix using only pointer arithmetic.

Input Format

You will receive the input matrix across three lines. Assume all values are integers.

Example:

> 1 2 -1
>
> 2 -1 3
>
> 3 1 2

Output Format

The calculated determinant.

Example:

> 0

2

When processing attention maps in a Transformer architecture, it is often necessary to reshape the dimensions of a 2D tensor (matrix) from $n \times m$ to $m \times n$. Write a C program that reads a matrix of $n \times m$ numbers, transposes the matrix, and then prints it out.

Instead of just printing the original matrix in a different order, you **must** perform the transpose operation by defining a second matrix in memory and copying the contents over.

Input Format

The first line contains n and m, the size of the feature matrix. The subsequent lines contain the contents of the matrix.

Example:

> 2 3
>
> 1 2 3
>
> 4 5 6

Output Format

The transposed matrix.

Example:

> 1 4
>
> 2 5
>
> 3 6

How to submit

Save your source code file in the form of `[StudentID]_A[Assignment Number]_P[Problem Number].c` . For example, for assignment 0522 problem 1, save it as `1W21CS08_A0522_P01.c` . Submit only the source code files.